

Pragnosia: Making a Diagonal-Complex Spin Recurrence the Load-Bearing Core of a Small Language Model

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ABSTRACT. We study a transformer/SSM hybrid in which a rotational "spin" recurrence — a diagonal-complex linear-recurrent unit run as a parallel associative scan — is the *core token-mixer*, with attention only a periodic helper. The work has a sharp, honest arc. We first scaled the hybrid *the wrong way*: a single spin carrier placed after a transformer stack, whose parameter share collapses $5.76\% \rightarrow 1.90\% \rightarrow 0.37\%$ from 21M to 1.4B; at 1.4B a causal probe shows the carrier contributes only **0.028% OF THE LOSS** — vestigial as built. A param-matched ablation then shows the *intended* design, spin as the mixer (`carrier="spin_dominant"`), reaches val perplexity **219 vs 279** for an attention-only baseline at matched parameters — spin earns its weights when it is the core. We validate this at scale for the first time: a **284M** spin-dominant model (val ppl 24.54 on a 176.6B-token corpus) in which *ablating the spin carrier sends perplexity* $25 \rightarrow 7716$ ($306\times$) while ablating attention costs only **3.4×** — the spin carrier is the load-bearing core, the exact inverse of the 1.4B drift. On this substrate runs a self-calibrating controller: continual learning by surprise-modulated plasticity with generative replay (no forgetting), an in-weights "subconscious" episodic memory, a metacognition/deliberation/monologue cognition loop, and function-preserving self-growth that fires only on probe-confirmed saturation. Every internal threshold is derived from data and re-derived as the model grows; nothing is hand-set. We report results and limits plainly: the spin-dominant win is small-scale, short-budget, single-seed and must be replicated at 1B; the grown models are undertrained for their size (multi-hop reasoning is weak); and the model's honesty signal, while functional, does not yet separate knowledge from confident confabulation at 284M.

1. Introduction

The dominant paradigm in AI is the large language model: a transformer¹ trained to predict the next token. LLMs are capable but, relative to the goal of a *brain*, three properties are missing by construction. They are *frozen* — once pretraining ends they cannot acquire a fact without fine-tuning or prompt-stuffing, and naive updates cause catastrophic forgetting². They are *over-confident* — lacking a model of their own uncertainty, they fabricate rather than abstain. And their computation is opaque — there is no standard intervention separating in-state computation from memorized retrieval.

Pragnosia is an attempt to build the missing properties into the substrate itself, even at small scale. Its contributions are: (i) a diagonal-complex rotational recurrence used as the *core token-mixer* via a parallel scan (§2); (ii) a measured analysis of *where that carrier earns its parameters* — our first scaled model drifted to attention-dominant with a causally-negligible carrier, while a param-matched ablation shows the intended *spin-dominant* design wins by ~21%, now confirmed load-bearing at 284M by ablation (§3); (iii) a self-calibrating controller with surprise-modulated continual learning, an in-weights episodic memory, a cognition loop, and function-preserving self-growth, with no hand-set constants (§4); and (iv) an honest account of what works and what does not (§5–7). We release the system and every measured number.

2. The Spin Carrier

A diagonal-complex recurrence, run as a parallel scan.

The carrier is a linear-recurrent unit⁸ over a complex hidden state. For input x (layer-normed to y), with an input-dependent gate $g=\sigma(G \cdot y)$, the per-channel recurrence and read-out are

$$\begin{aligned} b_t &= g \odot (U_{\text{re}} y_t + i \cdot U_{\text{im}} y_t), \quad h_t = \lambda \odot h_{t-1} + b_t, \\ \lambda_j &= \exp(-\exp(v_j)) \cdot e^{i\theta_j}, \quad \text{out}_t = C \cdot [\text{Re } h_t; \text{Im } h_t], \quad x_t \leftarrow x_t + \sigma(y) \cdot \text{out}_t. \end{aligned} \tag{2}$$

Each channel j is thus a *damped complex oscillator* at its own frequency θ_j and decay $|\lambda_j| \in (0, 1)$: information is carried in the evolving *phase* of the state, read back to the real stream through C and added by a learned gate $\sigma(y)$. Because λ is diagonal the recurrence is matmul-free — $O(\tau \cdot d)$, not $O(\tau \cdot d^2)$ — *and* associative, so it is computed not by a τ -step loop but by a log-depth **parallel associative scan** (a Hillis–Steele scan over the sequence). An in-place variant is exact and $\sim 20\%$ faster on the forward pass but breaks autograd; we keep the materializing scan. This is the only recurrence in the system; an earlier dense, tanh-nonlinear sequential variant is retained in version history but is not used.

Architecture and modes.

A SpinBlock uses the carrier as its token mixer (a strong gate, $\sigma(y) \approx 0.88$) followed by an MLP; a standard attention Block is the alternative. The model exposes one config flag, `carrier` $\in \{\text{none}, \text{single}, \text{per_block}, \text{spin_dominant}\}$, wired `pragnosia.json` $\rightarrow$ `train_pragnosia.py` $\rightarrow$ `grow.py`. `none` is a transformer; `single` places one carrier after the stack; `spin_dominant` — the intended design — makes the carrier the mixer in 3 of every 4 layers, with attention every 4th as a helper. At long context the carrier is $O(\tau \cdot d)$ versus attention's $O(\tau^2 \cdot d)$, and at inference it is recurrent: $O(1)$ per token, constant memory, no KV-cache growth. `torch.compile` fuses the scan for $\sim 2\times$ training throughput — measured $49K \rightarrow \sim 100K$ tok/s (bs 24) on a consumer RTX 4060. It is *not* faster than attention on short (256-token) training sequences; its wins are long context and recurrent inference.

3. Where the Spin Carrier Earns Its Parameters

The scaled model drifted to attention-dominant.

Our first scalable design placed a *single* gated spin carrier after a transformer stack. A single carrier is one fixed d -shaped layer ($\sim 5.25M$ params), so as the model grows by depth and width its *parameter share collapses*: 5.76% (21M) $\rightarrow 1.90\%$ (276M) $\rightarrow 0.37\%$ (1.4B). We measured its causal contribution by a *carried-state intervention* on the single-carrier model (transplant the carrier state across a segment boundary; a causal carrier shifts the continuation): at 1.4B the ordering `own < swapped < random` still holds *directionally*, but the contribution is only 0.00086 NLL — **0.028% of the loss**, down $\sim 20\times$ from 276M. As built, the carrier is *causally negligible at scale*: the model is a transformer with a vestigial side-channel. (We thank an external reviewer for flagging this.) The same 1.4B run also over-grew: an earlier trigger grew on any validation plateau rather than true saturation, so an undertrained model ballooned to 1.4B (48 layers, `mlp_mult` 12) long before the $\approx 28B$ tokens needed to fill it — over-growth, not a result. §4 gives the corrected growth rule.

The intended design is spin-dominant — and wins at matched parameters.

The drift inverts the intended architecture, where spin is the core mixer. We test the intended design with a clean param-matched ablation (same corpus, same budget, $d=512$, single seed):

Table 1. Param-matched ablation. Spin as the token mixer is the most parameter-efficient.

Design	Params	Val ppl
Attention-only (transformer)	35.7M	279.0
Spin-dominant (intended)	37.3M	219.1
Per-block hybrid (attn + spin each block)	46.2M	228.0
Param-matched transformer-only (11L)	45M	270

Spin-dominant wins by ~21% over the attention-only baseline at matched parameters and beats the larger per-block hybrid and param-matched transformer with fewer weights. **The carrier earns its parameters when it is the core mixer, not a side-channel.** This ablation is small-scale, short-budget, single-seed, and must be replicated at 200M–1B before it is proven; we report it as a strong, properly-controlled signal and the corrective to the drift.

At scale: the spin carrier is the load-bearing core (284M).

We trained the intended design at real scale for the first time. A **284M** model (d=768, 20 layers, mlp_mult=9, carrier="spin_dominant" — the carrier as mixer in 15 of 20 blocks, attention every 4th) trained to step 418,000 on a 176.6B-token chain-of-thought corpus, reaching **best validation perplexity 24.54** (≈ 24 tokens/param). In the spin-dominant configuration the carrier is *intra-block* with no cross-segment state to transplant, so the carried-state intervention above does not apply; the equivalent causal measure is direct *ablation*.

The carrier carries essentially the whole computation. Gating the 15 SpinBlocks' carrier to ≈ 0 sends validation perplexity from **25 to 7716** — **306 $\times$ worse**; zeroing the 5 attention blocks' output sends it from **25 to 85** — **3.4 $\times$ worse**. At 284M the spin carrier is the *load-bearing core* and attention a minor helper — the exact inverse of the 1.4B instance, whose attention-dominant-built carrier was vestigial (0.028% of the loss). The 284M is 5 $\times$ smaller than the 1.4B yet is the correct build with a provably load-bearing carrier.

4. A Self-Calibrating Living Controller

Around the language model runs a controller ("Pragnosia") that decides, for any input, whether to answer, abstain, learn, seek, or think — entirely from the model's own signals, with no keyword rules. Four design commitments distinguish it.

Nothing hardcoded.

Every internal scale is derived from data and re-derived by a `recalibrate()` step as the model grows. Word importance is the self-information $-\log p(\text{token})$ from the corpus (function words emerge low-weight, content words high, with no stopword list). Honesty, novelty, and memory-match thresholds are set at the equal-error-rate crossover of two of the model's *own* distributions (e.g. agreement on familiar text vs. agreement by chance), not hand-tuned. The training learning rate is found by a Smith range-test rather than seeded, and restored from a sidecar on resume; the growth bar is the data budget (Chinchilla ≈ 20 tokens/param) and VRAM, not a fixed number.

Surprise-modulated continual learning.

When taught a fact, the brain sets its own learning rate from its own surprise — prediction error relative to its familiarity boundary — learning hard on surprising input and gently on familiar input, and stopping once the fact ceases to surprise it. Teaching interleaves *generative replay* of the training distribution so a new fact does not erase old skills; in practice taught facts persist across sessions and prior abilities are retained (no catastrophic forgetting). An in-weights **subconscious** episodic memory complements this: it writes surprise-gated key $\rightarrow$ value traces of the model's own activations, recalls them under an *uncertainty gate* (so it never contaminates confident

outputs), decays/forgets over time, and consolidates into the slow weights during a "sleep" step. It is a learned in-weights store, **not** retrieval or RAG — there is no external index at recall time.

Cognition and self-growth.

A cognition loop adds metacognition (the model judging its own confidence), step-by-step deliberation (an internal scratchpad), and an autonomous internal monologue when idle. Growth is function-preserving and fires only on *probe-confirmed saturation*: the model spawns a depth-grown and a width-grown copy, trains each briefly, and commits to growth only if validation loss drops past the validation noise — choosing depth vs. width by whichever helps more, capped by the derived data budget and VRAM. Because the diagonal carrier is d-shaped, growth composes with it directly. This rule is confirmed in production: a spin-dominant run grew itself 24M (4 layers) → 27.3M (5 layers) on confirmed saturation, then stopped on its own at the data budget — the correction to the earlier over-growth to 1.4B.

5. Experiments and Results

Trained instances.

The diagonal-complex carrier trains stably from 21M to 1.4B. Two instances anchor the analysis of §3: the self-grown **1.4B** single-carrier model (val ppl 17.35 after adaptive-lr refinement) is the attention-dominant "wrong build" — bigger, but with a causally-negligible carrier; the **284M** spin-dominant model (val ppl 24.54) is the corrected build, 5× smaller, with a provably load-bearing carrier. At these scales the models are coherent at the sentence level, write simple correct code (`def add(a,b): → return a + b`), and produce the *form* of step-by-step reasoning; single-step facts and arithmetic are real while multi-hop composition is weak — the undertraining-for-size signature, expected before the compute-optimal token budget is reached. The training corpus (`window2_train`) is 330 GB / 176.6B tokens — 158.6B of base text plus ~18B of appended chain-of-thought reasoning, targeting that weak axis.

Capability and honesty of the 284M model.

A greedy capability battery on the 284M shows a clear jump from a 27M laptop model (27M in parentheses): arithmetic **3/3** (0/3), knowledge **2/3** (0/3), logic **2/2**, theory-of-mind **2/2** (1/2), code 1/2, in-context binding 1/2; the still-weak axis is multi-step 0/2, counterfactual 0/2, causal 0/2. Sample generations: "The capital of France is → the city of Paris.", " $2 + 2 = → 4$.", "Once upon a time →, in a small town named Harmonyville, lived two best friends". World knowledge, single-step arithmetic, and theory-of-mind genuinely *emerged* at this scale; multi-step composition is the next axis.

An honest negative on honesty. The controller answers only when its sampled answers *agree* on a content token (real knowledge pins one answer; confabulation varies). Asked in the model's trained chat format this works for some facts, but at 284M the signal does *not* cleanly separate knowledge from confident confabulation: a known fact ("capital of France") and an unknowable ("my secret password", "who wins the 2031 election") score 0.53 / 0.53 / 0.67 — overlapping. The model confabulates *consistently*. The gate errs toward "I don't know" rather than guessing; reliable honesty needs either scale (consistency sharpens with model size) or a different confidence signal. We do not claim the chat form is reliably honest.

6. Comparison with Large Language Models

Table 2. Where each paradigm wins. The two are built on opposite bets.

Property	Today's LLMs	Pragnosia
Fluent language / world knowledge	Strong / vast	Decent / limited (small)
Reasoning depth	Strong	Single-step real, multi-hop weak
Long-context / inference cost	$O(T^2)$, KV-cache grows	$O(T \cdot d)$, recurrent $O(1)$ /token
Learn after training, no retrain	No (frozen)	Yes (replay + plasticity)
Forgetting on update	Catastrophic	Low (generative replay)
Self-paced learning rate	Fixed schedule	Surprise-modulated
Hand-set behavior	—	None (self-derived)
Knows what it doesn't know	Partial, bolted on	Abstains, but signal scale-limited
Size / cost	Data-center	Single laptop GPU

7. Limitations

Every claim above carries a number; the open frontier is equally explicit. (i) **The at-scale evidence is single-seed and not yet at 1B.** The ~21% param-efficiency margin is shown at ~37M and the load-bearing causality at 284M; multi-seed replication and a matched 1B run — including a carrier="none" transformer baseline at 284M, which proves the *win* rather than only that the carrier is load-bearing — remain to fully establish the design. (ii) **The models are undertrained for their grown size.** Single-step reasoning, arithmetic, and fact recall are real, but multi-hop composition — relational analogy, multi-step deduction, counterfactual and causal chains — is weak. The lever is more reasoning-rich tokens, not more parameters or prompting. (iii) **Honesty is not solved.** The consistency-based gate abstains on the clearly-unknowable but cannot separate knowledge from confident confabulation at 284M (§5); it needs scale or a redesigned signal. (iv) Like all small models it can be *confidently wrong* on familiar-but-incorrect facts. None of these is hidden.

8. Conclusion

Pragnosia is a transformer/SSM hybrid in which a diagonal-complex spin recurrence is the core token-mixer, wrapped in a self-calibrating controller for continual learning, an in-weights episodic memory, a cognition loop, and self-governed growth — with no hand-set constants. The central, honest finding is about *placement*: a single spin carrier drifts to a causally-negligible side-channel as the model scales (0.028% of the loss at 1.4B), and a param-matched ablation shows the intended spin-dominant design — spin as the core mixer — is the most parameter-efficient (219 vs 279 ppl). At 284M, the first at-scale training of the intended build, ablating the carrier costs 306× in perplexity versus 3.4× for attention: the spin carrier is the load-bearing core, the exact inverse of the 1.4B drift. The remaining work is a matched transformer baseline and multi-seed replication at 1B, and training the models to their compute-optimal token budget. We release the full system, every measured number, and every limitation.

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